

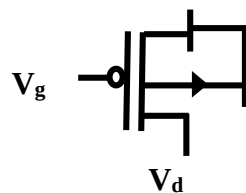
In this lab we will

- Analyze the I-V characteristics of the NMOS and PMOS transistors using LTspice.
- Design a transmission gate
- Test the ability of the NMOS and PMOS transistors to transfer voltage.

PART 1: Transistor Analysis

Build the circuit shown in the figure below using **PMOS** transistor and answer the following questions:

$$V_s = V_{DD} = 1.2V$$



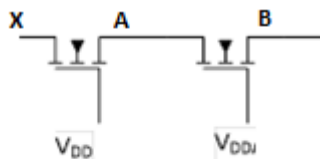
- 1- For $W=200n$ and $L=50n$ find the drain current I_d and the mode of operation when the gate voltage is $0.3V$ and the drain voltage is $0.2V$.
- 2- Draw the I-V characteristics (V_{ds} vs I_{ds}) for the PMOS transistor for the following gate voltage values ($0, 0.3, 0.6$ and 0.8)
- 3- Draw the relation between the Length of the transistor and the current for the values used in Q1.
- 4- Draw the relation between the temperature and the current for the values used in Q1.
- 5- **[Post Lab]** Draw the relation between the body voltage and the current for the values used in Q1.

PART 2: Pass transistors and Transmission Gate

- 1- Using pass transistor design an experiment to measure the V_{th} voltage of an NMOS transistor.

Hint: Connect the gate of the transistor to V_{dd} and apply a pulse signal at node X.

- 2- Draw the circuit below to find the voltage at Node A and B in the figure below.



PART 3: Transmission Gate(Post Lab)

Design a transmission gate using NMOS and PMOS transistors and then design an experiment to test its ability to transfer strong 0 and strong 1.